

Problem 1 (25%)

A box contains 5 red balls, 4 blue balls, and 3 green balls. Three balls are drawn sequentially without replacement.

Let

$$A = \{\text{the first ball is red}\},$$

and let

$$C = \{\text{at least one of the three drawn balls is red}\}.$$

Suppose that the only information revealed after the experiment is that event C occurred; that is, at least one of the three drawn balls was red.

- (a) Find $P(C)$.
- (b) Find $P(A \cap C)$ and hence calculate $P(A | C)$.
- (c) Let B be the event that the second ball is red. Calculate $P(B | C)$.
- (d) Calculate $P(A \cap B | C)$.
- (e) Compare $P(A | C)$ with $P(A)$. Are A and C independent? Justify your answer using the definition of independence.

Solution

(a)

complement of C : none of the three balls is red

$$P(C^c) = \frac{7}{12} \cdot \frac{6}{11} \cdot \frac{5}{10} = \frac{7}{44}.$$

$$\implies P(C) = 1 - P(C^c) = 1 - \frac{7}{44} = \frac{37}{44}$$

(b)

$$\because A \subseteq C,$$

$$\therefore A \cap C = A.$$

$$\implies P(A \cap C) = P(A) = \frac{5}{12}$$

$$\implies P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{5}{12}}{\frac{37}{44}} = \frac{55}{111}$$

(c)

Let

$$B = \{\text{the second ball is red}\}.$$

$$\because B \subseteq C,$$

$$\therefore P(B \cap C) = P(B).$$

By symmetry,

$$P(B) = \frac{5}{12}.$$

$$\implies P(B | C) = \frac{P(B \cap C)}{P(C)} = \frac{\frac{5}{12}}{\frac{37}{44}} = \frac{55}{111}$$

(d)

$$\because P(A \cap B) = \frac{5}{12} \cdot \frac{4}{11} = \frac{5}{33}.$$

$$\therefore A \cap B \subseteq C,$$

$$\therefore (A \cap B) \cap C = A \cap B.$$

$$\implies P(A \cap B | C) = \frac{P(A \cap B)}{P(C)} = \frac{\frac{5}{33}}{\frac{37}{44}} = \frac{20}{111}$$

(e)

$$\because P(A) = \frac{5}{12},$$

while

$$P(A | C) = \frac{55}{111}.$$

$$\therefore \frac{55}{111} \neq \frac{5}{12},$$

$$\implies P(A | C) \neq P(A).$$

By the definition of independence,

$$A \text{ and } C \text{ are independent} \iff P(A \cap C) = P(A)P(C).$$

$$\because P(A \cap C) = \frac{5}{12},$$

$$P(A)P(C) = \frac{5}{12} \cdot \frac{37}{44} = \frac{185}{528}.$$

$$\implies \frac{5}{12} = \frac{220}{528} \neq \frac{185}{528},$$

$\therefore A$ and C are not independent

Problem 2 (25%)

Multiple Detection System

A monitoring system uses four independent sensors to detect whether a machine has experienced a failure. Let A_i denote the event that sensor i raises a false alarm. The false-alarm probabilities are

$$P(A_1) = 0.10, \quad P(A_2) = 0.15, \quad P(A_3) = 0.20, \quad P(A_4) = 0.25.$$

Assume that the four events are mutually independent.

The monitoring system issues a warning if at least one sensor raises a false alarm.

- (a) Suppose the monitoring system has issued a warning. Find the probability that at least two sensors have raised a false alarm, i.e.,

$$P(\text{at least two alarms} \mid A_1 \cup A_2 \cup A_3 \cup A_4).$$

- (b) Suppose now that no independence assumption is made. Instead, you are given

$$P(A_i) = 0.1, \quad i = 1, 2, 3, 4,$$

and

$$P(A_i \cap A_j) = 0.02$$

for every $i \neq j$, together with

$$P(A_i \cap A_j \cap A_k) = 0.005$$

for every distinct i, j, k , while

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = 0.001.$$

Use the inclusion-exclusion principle to calculate

$$P(A_1 \cup A_2 \cup A_3 \cup A_4).$$

Solution

(a) Let

$$W = A_1 \cup A_2 \cup A_3 \cup A_4$$

be the event that the system issues a warning, and let

$$B = \{\text{at least two sensors raise false alarms}\}.$$

Since $B \subseteq W$,

$$P(B | W) = \frac{P(B \cap W)}{P(W)} = \frac{P(B)}{P(W)}.$$

First,

$$P(W) = 1 - P(A_1^c \cap A_2^c \cap A_3^c \cap A_4^c).$$

By independence,

$$\begin{aligned} P(W) &= 1 - (0.90)(0.85)(0.80)(0.75) \\ &= 1 - 0.459 \\ &= 0.541. \end{aligned}$$

Next,

$$P(B) = 1 - P(\text{no alarm}) - P(\text{exactly one alarm}).$$

$$\therefore P(\text{no alarm}) = (0.90)(0.85)(0.80)(0.75) = 0.459.$$

The probability of exactly one alarm is

$$\begin{aligned} P(\text{exactly one alarm}) &= (0.10)(0.85)(0.80)(0.75) \\ &\quad + (0.90)(0.15)(0.80)(0.75) \\ &\quad + (0.90)(0.85)(0.20)(0.75) \\ &\quad + (0.90)(0.85)(0.80)(0.25) \\ &= 0.051 + 0.081 + 0.11475 + 0.153 \\ &= 0.39975 \end{aligned}$$

$$\therefore \begin{aligned} P(B) &= 1 - 0.459 - 0.39975 \\ &= 0.14125. \end{aligned}$$

$$\therefore \begin{aligned} P(B | W) &= \frac{0.14125}{0.541} \\ &\approx 0.2611. \end{aligned}$$

$$\implies P(\text{at least two alarms} | A_1 \cup A_2 \cup A_3 \cup A_4) \approx 0.2611$$

(b) By the inclusion-exclusion principle,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) &= \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) \\ &\quad + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) \\ &\quad - P(A_1 \cap A_2 \cap A_3 \cap A_4). \end{aligned}$$

There are

$$\binom{4}{1} = 4$$

single events,

$$\binom{4}{2} = 6$$

pairwise intersections, and

$$\binom{4}{3} = 4$$

triple intersections.

Therefore,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) &= 4(0.1) - 6(0.02) + 4(0.005) - 0.001 \\ &= 0.4 - 0.12 + 0.02 - 0.001 \\ &= 0.299. \end{aligned}$$

$$\implies P(A_1 \cup A_2 \cup A_3 \cup A_4) = 0.299$$

Problem3

A factory has three production lines, L_1 , L_2 , and L_3 , which produce 30%, 45%, and 25% of the factory's total products, respectively. The probabilities that a product is defective given that it comes from L_1 , L_2 , or L_3 are

$$P(D \mid L_1) = 0.02, \quad P(D \mid L_2) = 0.03, \quad P(D \mid L_3) = 0.05,$$

where D denotes the event that a randomly selected product is defective.

- Two products are selected independently from the factory. Find the probability that both products are defective.
- Three products are selected independently from the factory. Find the probability that at least one of the three products is defective.
- Suppose n products are selected independently from the factory. Find the smallest integer n such that the probability of obtaining at least one defective product is at least 95%.

Solution

by the law of total probability,

$$\begin{aligned}P(D) &= P(D \mid L_1)P(L_1) + P(D \mid L_2)P(L_2) + P(D \mid L_3)P(L_3) \\&= (0.02)(0.30) + (0.03)(0.45) + (0.05)(0.25) \\&= 0.006 + 0.0135 + 0.0125 \\&= 0.032.\end{aligned}$$

$$\therefore P(D) = 0.032, \quad P(D^c) = 1 - 0.032 = 0.968.$$

(a)

$\therefore$ the two products are selected independently

$$\therefore P(\text{both defective}) = P(D)^2 = (0.032)^2 = 0.001024.$$

$$\implies P(\text{both defective}) = 0.001024$$

(b)

The event “at least one product is defective” is the complement of “none of the three products is defective.” $\therefore$ the selections are independent,

$$P(\text{none defective}) = (0.968)^3.$$

$$\begin{aligned}\therefore P(\text{at least one defective}) &= 1 - P(\text{none defective}) \\&= 1 - (0.968)^3 \\&= 0.092960768.\end{aligned}$$

$$\implies P(\text{at least one defective}) \approx 0.0930$$

(c)

For n independently selected products,

$$P(\text{no defective products}) = (0.968)^n.$$

$$\therefore P(\text{at least one defective product}) = 1 - (0.968)^n.$$

$$\because 1 - (0.968)^n \geq 0.95.$$

$$\therefore (0.968)^n \leq 0.05.$$

Taking natural logarithms,

$$n \ln(0.968) \leq \ln(0.05).$$

Since

$$\ln(0.968) < 0,$$

dividing by $\ln(0.968)$ reverses the inequality:

$$n \geq \frac{\ln(0.05)}{\ln(0.968)}.$$

Hence,

$$n \geq 92.11 \dots$$

$\implies n$ must be an integer, the smallest possible value is

$$n = 93$$

Problem4

Two dice are rolled independently. For each die, the probability of showing an even number is p , where

$$0 < p < 1.$$

The two dice have the same distribution, but no assumption is made that the individual outcomes $1, 2, \dots, 6$ are equally likely.

Define

$$A = \{\text{the sum of the two dice is even}\},$$

and

$$B = \{\text{the sum of the two dice is greater than 8}\}.$$

Determine whether there exists a value of p such that, for every possible distribution of the individual die outcomes satisfying

$$P(\text{even}) = p,$$

the events A and B are independent.

If such a value of p exists, find all such values and prove that the independence holds for every admissible distribution.

If no such value exists, prove this by constructing, for each candidate value of p , two admissible distributions that have the same value of p but give different values of

$$P(A \cap B) - P(A)P(B).$$

Your argument should distinguish carefully between independence that is implied by the given information and independence that occurs only for a particular choice of the underlying distribution.

Solution

Let the two dice outcomes be X and Y .

$$P(X \text{ is even}) = P(Y \text{ is even}) = p,$$

the two dice are independent

$$\implies P(X \text{ is odd}) = P(Y \text{ is odd}) = 1 - p.$$

The event A occurs if both dice are even or both dice are odd

$$\implies P(A) = p^2 + (1 - p)^2.$$

Thus $P(A)$ depends only on p .

We now construct two admissible distributions having the same value of p .

Distribution 1

Let

$$P(X = 1) = 1 - p, \quad P(X = 2) = p,$$

and

$$P(X = k) = 0, \quad k = 3, 4, 5, 6.$$

The same distribution is used for Y .

Clearly,

$$P(\text{even}) = p.$$

The largest possible sum is

$$2 + 2 = 4,$$

so

$$P(B) = 0.$$

$$\therefore P(A \cap B) = 0,$$

$$\therefore P(A \cap B) - P(A)P(B) = 0.$$

Thus, for this particular distribution,

A and B are independent.

Distribution 2

Now let

$$P(X = 1) = 1 - p, \quad P(X = 6) = p,$$

and

$$P(X = k) = 0, \quad k = 2, 3, 4, 5.$$

Again, Y has the same distribution.

$$\therefore P(\text{even}) = p.$$

$\therefore$ The possible sums are

$$2, \quad 7, \quad 12.$$

The event B occurs only when

$$X = Y = 6.$$

Therefore

$$P(B) = p^2.$$

Also, $6 + 6 = 12$ is even, so

$$B \subseteq A.$$

Hence

$$P(A \cap B) = P(B) = p^2.$$

Therefore

$$\begin{aligned} P(A \cap B) - P(A)P(B) &= p^2 - [p^2 + (1 - p)^2] p^2 \\ &= p^2 [1 - p^2 - (1 - p)^2] \\ &= p^2 [2p - 2p^2] \\ &= 2p^3(1 - p). \end{aligned}$$

Since

$$0 < p < 1,$$

we have

$$2p^3(1 - p) > 0.$$

Thus

$$P(A \cap B) \neq P(A)P(B),$$

so A and B are not independent for this distribution.

The two distributions have the same value of

$$P(\text{even}) = p,$$

but

$$P(A \cap B) - P(A)P(B) = 0$$

for the first distribution, whereas

$$P(A \cap B) - P(A)P(B) = 2p^3(1 - p) > 0$$

for the second distribution.

Therefore, for every $p \in (0, 1)$, there exists an admissible distribution for which A and B are not independent.

$\implies$ there is no $p \in (0, 1)$ such that A and B are independent for every admissible distribution.

The independence in the first construction occurs only because of that particular underlying distribution; it is not implied by the condition

$$P(\text{even}) = p.$$